UNIVERSITY OF AUCKLAND
Waipapa Taumata Rau
NEW ZEALAND

SCIENCE

COMPSCI 714
Lectutorial 1: Data basics

Data loading, exploring, cleaning and pre-processing

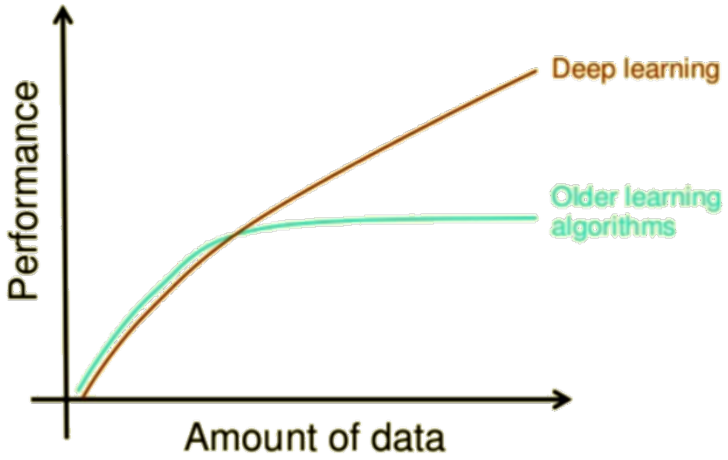
Instructor: Thomas Lacombe

Learning outcomes

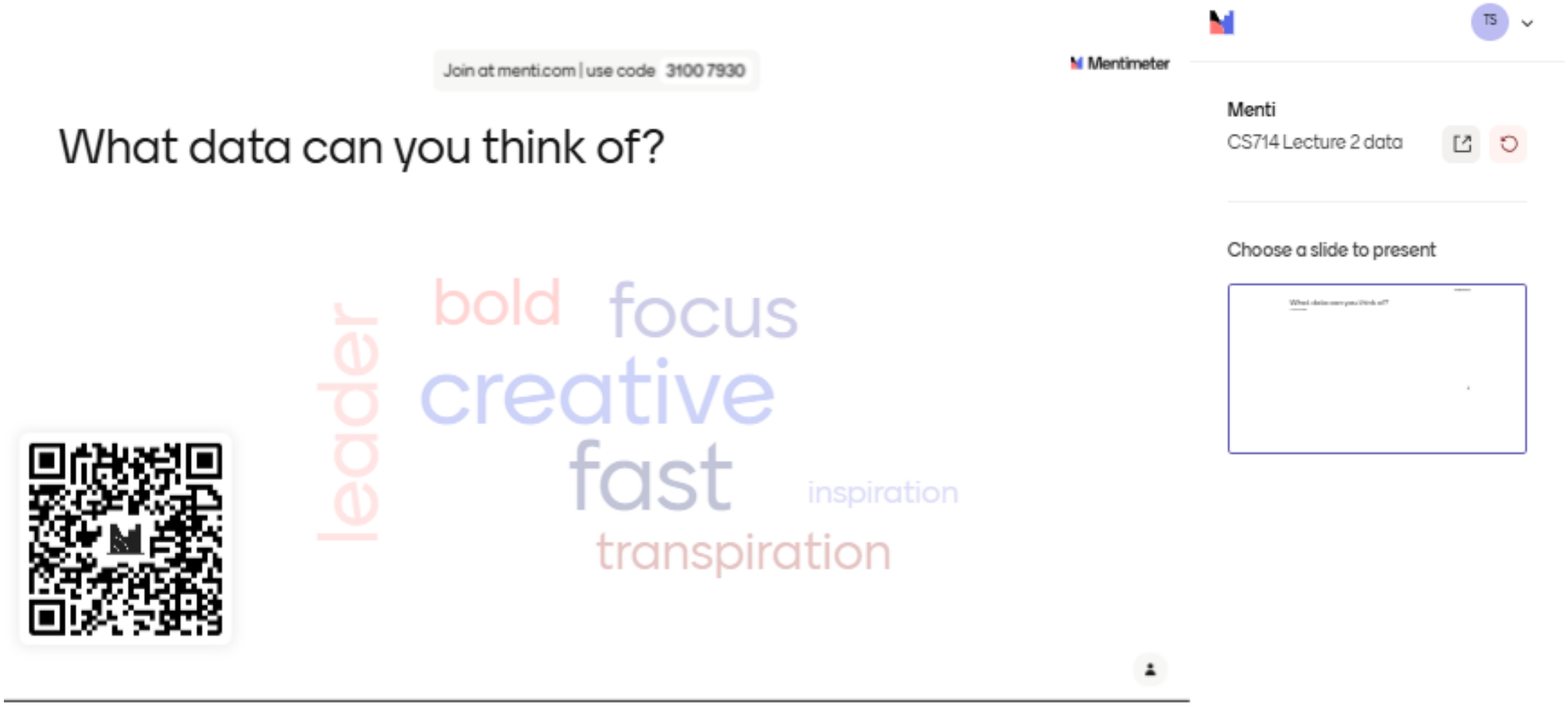
- Understand the role of data in the development of modern AI systems.
- Identify different sources of data.
- Understand processes and considerations involved when acquiring and using data to train a model.
- Understand the importance of exploring the data before starting to train a model.
- Identify different cleaning and pre-processing steps to prepare the data for training a model.

Data is fuel for AI and Machine Learning systems

- Training models with a lot of parameters (i.e., degrees of freedom) requires a lot of data.
- Common « 10 times » rule for deep neural networks.
- High volume of data collected via the internet: « Big data »



What is data?


Join at menti.com | use code 3100 7930
What data can you think of?
leader
bold
focus
creative
fast
inspiration
transpiration
Mentimeter
TS
Menti
CS714 Lecture 2 data
Choose a slide to present

What is data?

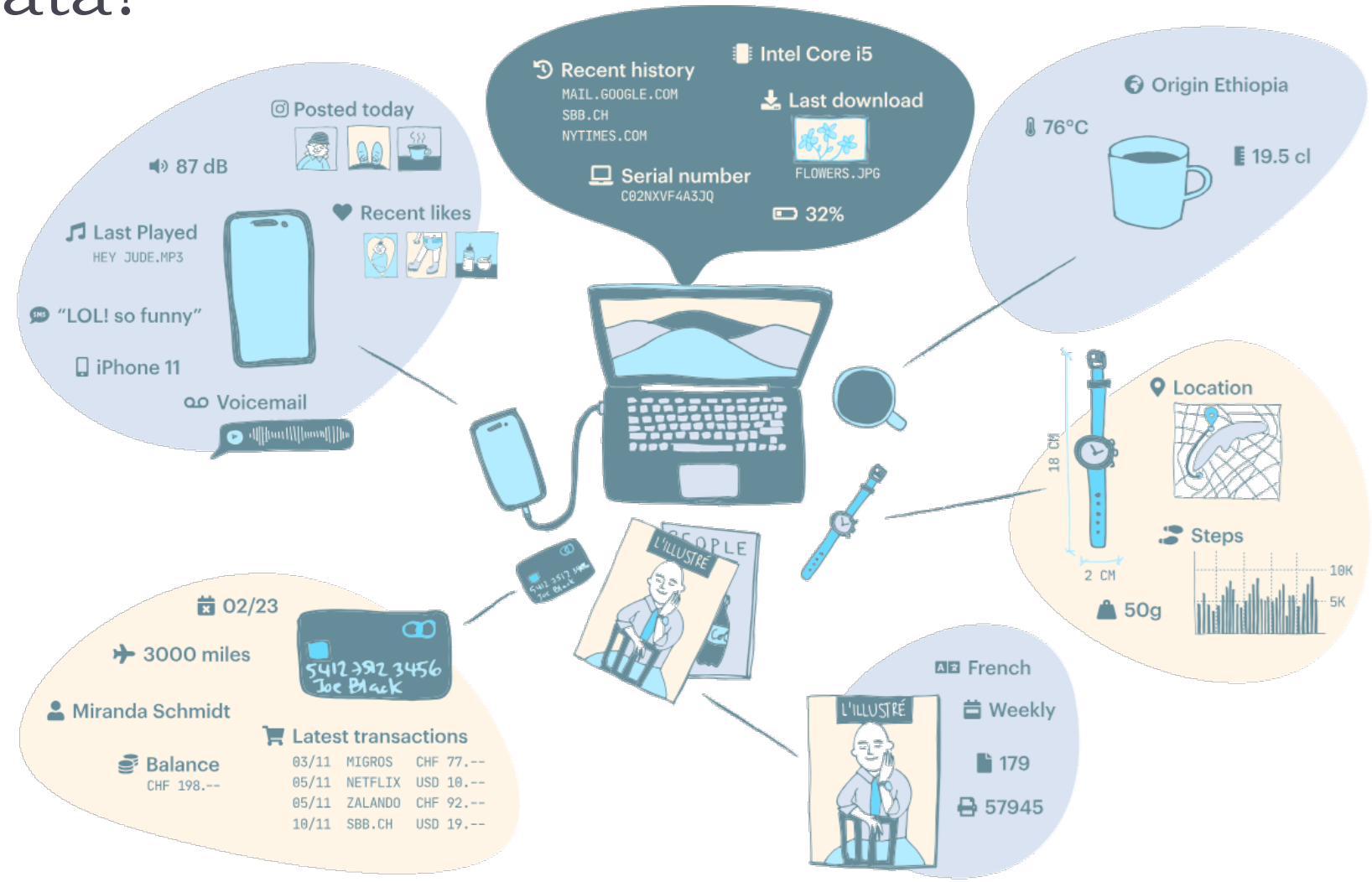


Image Source: https://www.thats-ai.org/en-GB/units/data-comes-in-many-forms

What is data?

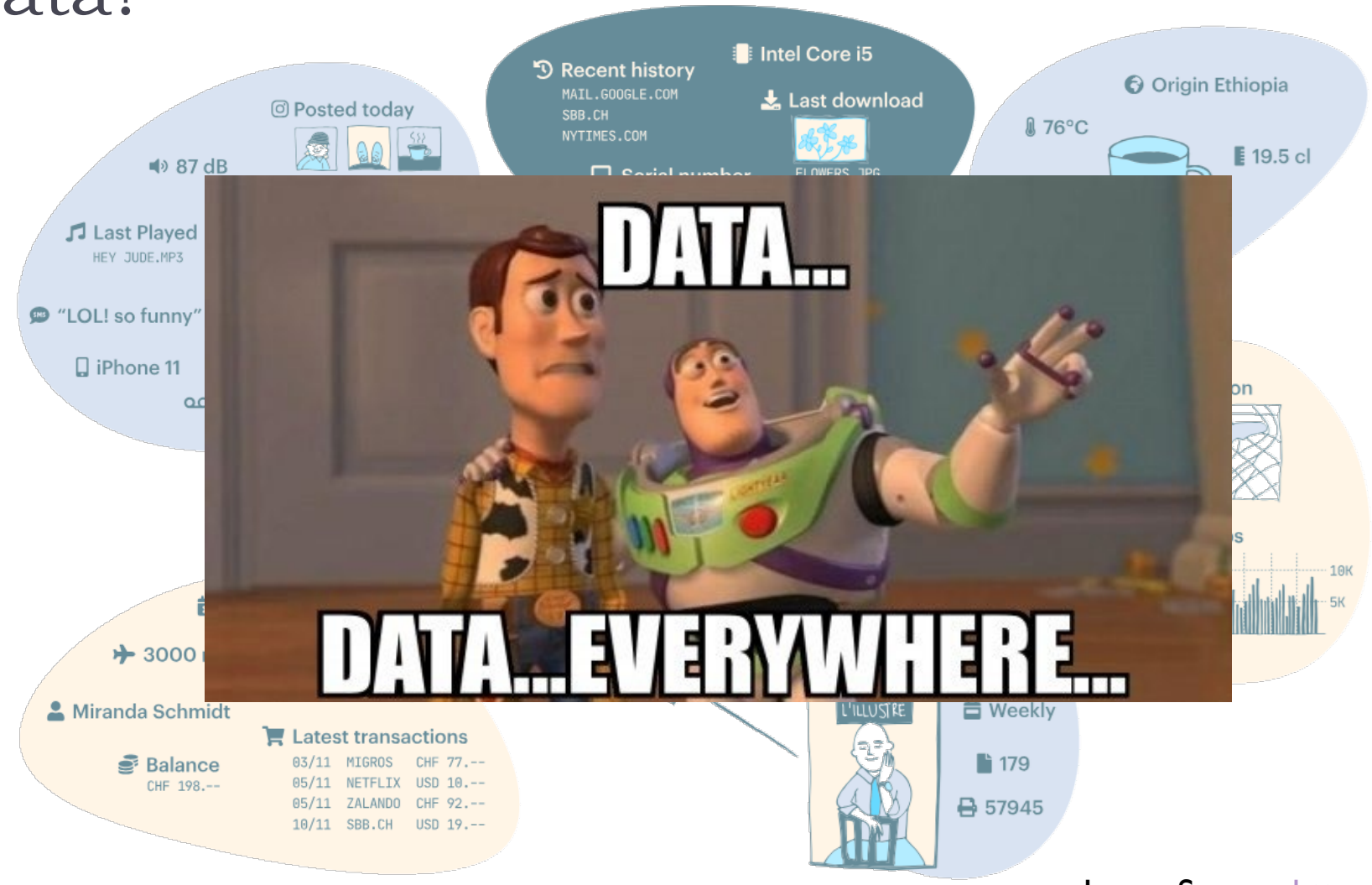


Image Source: https://www.thats-ai.org/en-GB/units/data-comes-in-many-forms

Forms of data

- Data can come in various forms:

Form of data	Structured data	Unstructured data
What is it?	Data following a standard format or data model, typically tabular with columns defining data attributes and rows containing different instances of data.	Data not following a specific format or data model.
Examples	Excel table, SQL databases.	Collection of image/video/audio files, emails, social media posts.
Typical storage method	Various types of structured databases (e.g., relational).	File systems.

- Structured data is typically easier to organise and manipulate (e.g., clean, search).
- Not all data can be shaped into a structured form.

Example of structured tabular data

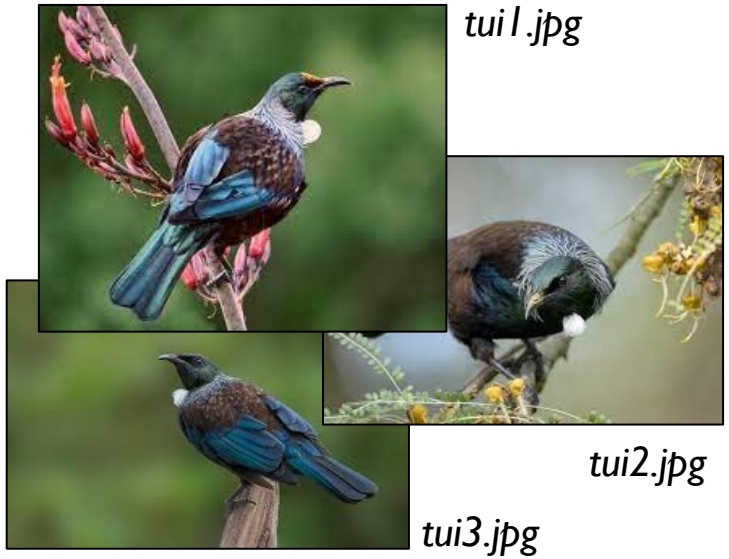
Index	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200.0	NEAR BAY

Loading data and basic exploration

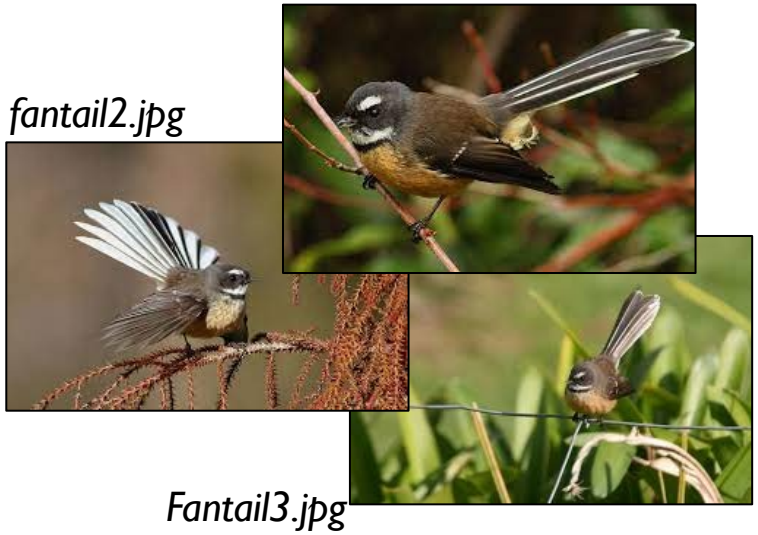
Types of attributes

- Data attributes in tabular data can have 2 main types:
  - Numerical: typically used to store quantitative data
    - Continuous (can take any numeric value, e.g., age)
    - Discrete (can take only a finite number of specific values, e.g., number of own cars)
  - Categorical: typically used to store quantitative data
    - Nominal (data can fit different categories, e.g., animals being mammal, fish, reptile, etc.)
    - Ordinal (data fits in ranked/ordered categories, e.g., bad/average/good/excellent)
    - Binary (two options, e.g., True/False)

Example of unstructured data



Folder: Tui



Folder: Fantail

Folder: Kiwi

Where can you get data?

- Repositories (e.g., Kaggle, GitHub, OpenML, HugginFace, etc.)
- Python Packages (Scikit-Learn datasets, PyTorch/TensorFlow datasets, etc.)
- Your own data (you will need to shape it yourself)
- Web scrapping

→**Important: always check if you can use the data (rights, confidential attributes, etc.) and if you need to take any precautions (e.g., anonymisation).**

→**I will share a Notebook about personal data you can have a look at as extra material.**

Dataset characteristics

- After you load a dataset, start by looking at its global characteristics:
  - What is the type of data? (e.g., tabular, images, text, …) / How is it organised?
  - What is the type of the target? (i.e., is it a classification or regression task?)
  - *Classification*: How many classes are they? / *Regression*: What is the target's range of values?
  - How many instances are there in total?
  - *Classification*: How many instances are there per class? (i.e., is there some class imbalance?)
  - What is the type of each attribute?
  - How many attributes are there?

Data imbalance

- Data imbalance happens when one or several classes are overrepresented (majority classes) in the data, while some other classes are underrepresented (minority classes).
- Examples:
  - *Unbalance in the target classes*: you want to classify images into 2 classes, cat and dog. Your dataset has 100 000 images in total, 95000 images of cats and 5000 image dogs.
  - *Unbalance in a predictive attribute*: you want to predict the probability of somebody getting a loan based on their age, gender, salary, etc. and you have 95% of your data with gender = male.

Degree of imbalance	Proportion of Minority Class
Mild	20-40% of the data set
Moderate	1-20% of the data set
Extreme	<1% of the data set

Table source: Google ML course

Why and when is it a problem?

- Many classification and regression problems are imbalanced! It is hard, sometimes impossible, to collect balanced data (e.g., fraud detection).

- When it can be a problem:
  - When splitting your data in different sets (e.g., train, development, test sets):
    You need to keep the same distributions for the target and important attributes in your different sets.
  - When training a model:
    Your model might become very good at predicting the majority class(es), but not the minority class(es).
  - When getting insights from a trained model:
    Because of the previous point, the insight you get from your model once trained might not reflect the actual patterns.

How to alleviate imbalance

- There exists several strategies to alleviate imbalance depending on the situation:
  - When splitting your data in different sets (e.g., train, development, test sets):

    Use special sampling approaches such as stratified sampling to sample the sets while maintaining the data distribution.
  - When training a model:

    Use appropriate “imbalance aware” metrics such as precision and recall to evaluate the performance of your model (we will cover that in a future lecture).
  - Acting directly on the data:

    Use upsampling or downsampling, possibly combined. However, these should be used carefully as you might end up losing important information or introducing unwanted bias. Not recommended in small datasets.

Are you sure your data contains data?

- Remember that machine learning algorithms are trying to build models based on statistical patterns in the data, and relationship between inputs and outputs.

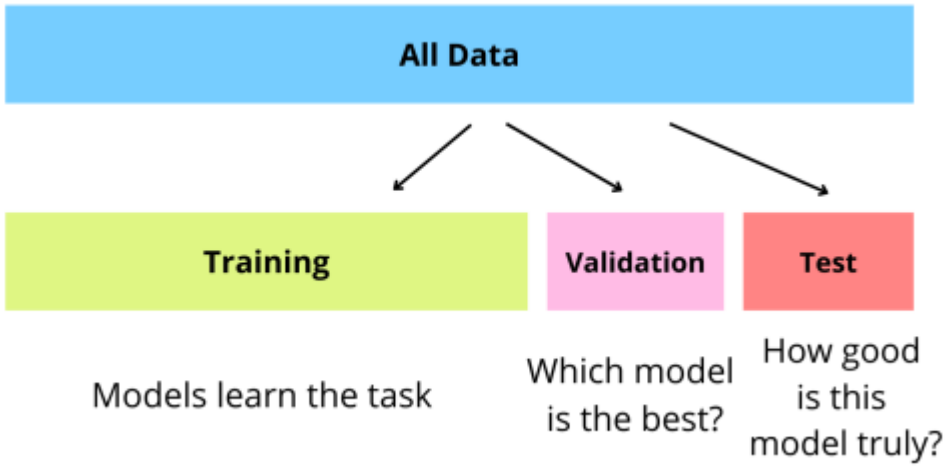
- There might be situations where your data does not contain much “signal” to be used to build a model to answer a question. Therefore, before starting to train a model, you should answer the question "Does my data contain the potential to answer this question?"

- For example:
  - Predicting Completely Random Events (e.g., model the outcome of a coin toss)
  - Inscrutable tasks (e.g., some human behaviour can be very hard to model, even with tons of data)
  - Insufficiently Related Data (e.g., predicting NFT price based on image features)

Before exploring the data further

- To answer the previous question, you need to explore the data further.

- But before this, you need to split your data.



- Very important to do the exploration only on the training set, to avoid data leakage.

Splitting your data in train and test sets & Further exploration

Exploring the data further

- After having a look at the overall characteristics of the data, you can push the exploration further:

Type of data	Examples of possible actions
Tabular numerical data	• Display scatter plots, histograms, etc., to visualise your data (can use some dimensionality reduction methods if your data has a lot of attributes/dimensions) • Display correlation matrices and plots to visualise relationship between data attributes • Try attribute combinations
Tabular categorical data	• Display the different categories to identify if they are nominal or ordinal
Images	• Display some images for each class • Calculate image sizes and aspect ratios
Text	• Display some text samples • Calculate some text statistics and visualise them with histograms • Depending on the task, you could perform a sentiment analysis, display a word cloud, etc.

- A good understanding of your data is useful to make cleaning and pre-processing choices and to guide the design of your model and training process.

Data cleaning

- One of the most time-consuming task when preparing data is to clean it.

- Cleaning steps may involve:
  - Removing duplicates (e.g., duplicate images)
  - Fixing formatting issues (e.g., inconsistent floating-point formatting)
  - Dealing with missing values
  - Dealing with outliers

- You can find a more detailed checklist on datacamp.

- Widely used datasets are usually already cleaned when you load them.

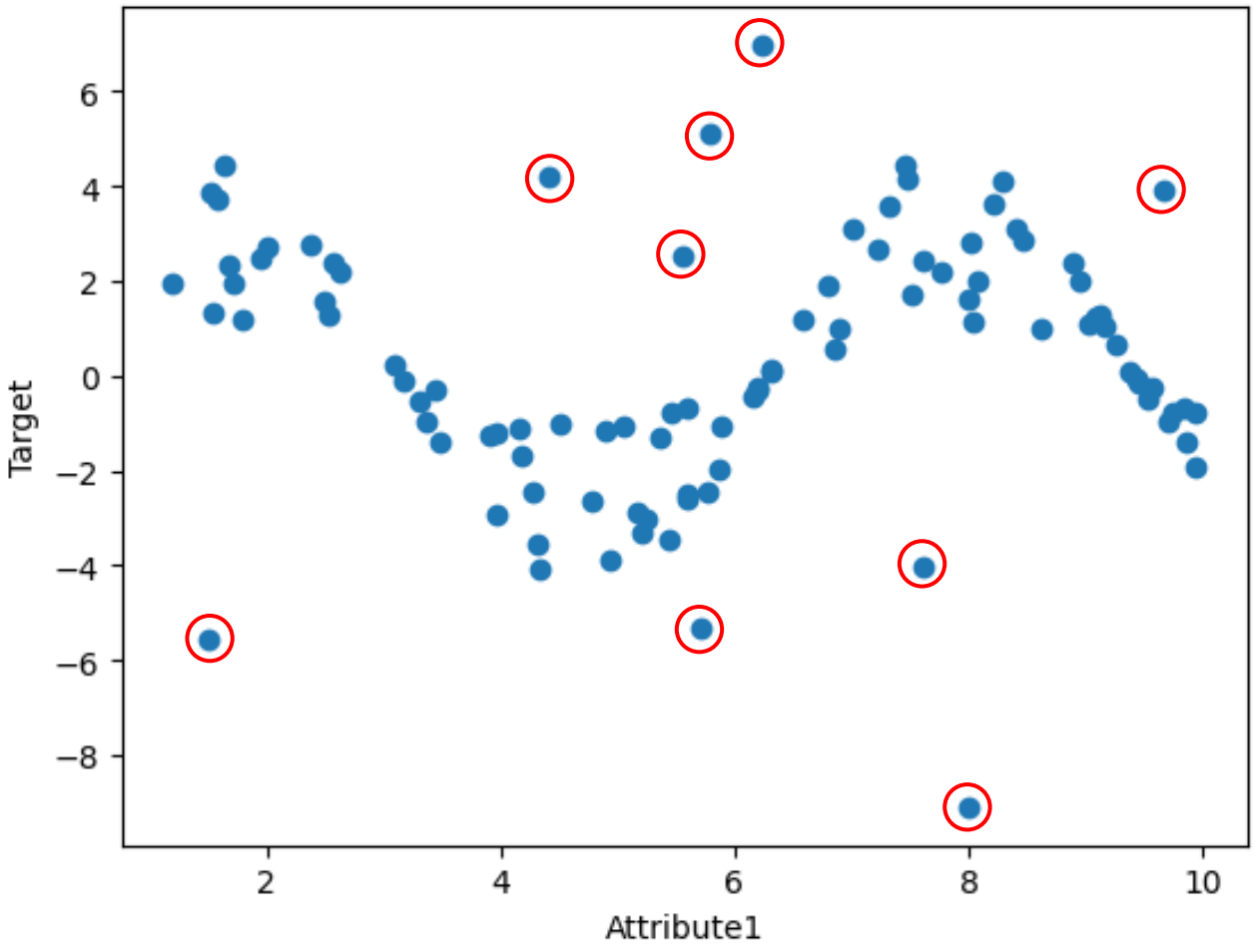
Missing values

Index	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
0	-122.23	37.88	41.0	880.0	129.0	322.0	**NaN**	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	**NaN**	7099.0	1106.0	2401.0	**NaN**	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	**NaN**	7.2574	352100.0	NEAR BAY
3	-122.25	**NaN**	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	-122.25	37.85	52.0	1627.0	280.0	565.0	**NaN**	3.8462	342200.0	NEAR BAY

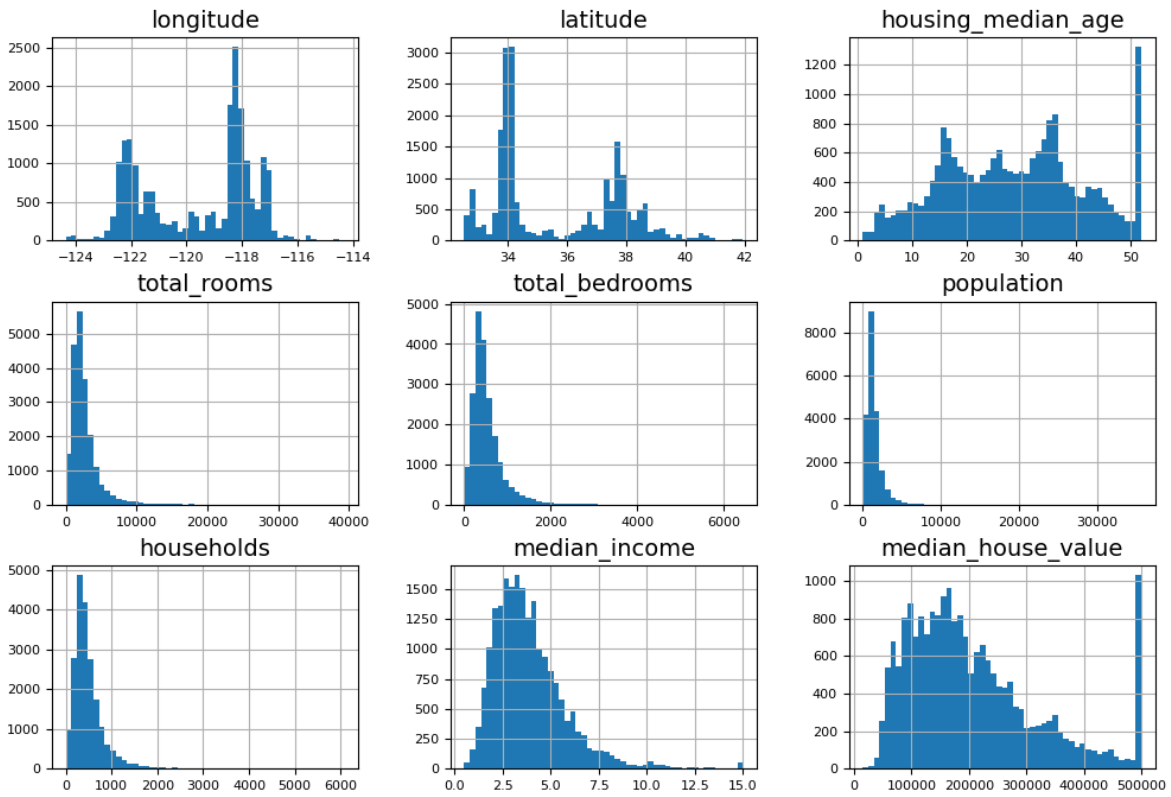
- Possible strategies involve removing an instance (row) or a full attribute (column), or use imputation methods to replace the missing values.

Outliers

- Outliers are data points which are « out of pattern ».
- Can be due to measurement error, data corruption, or can be true outliers.
- Not part of the pattern your model should describe.
- There are a lot of ways to detect outliers, including statistical and ML approaches (e.g., Isolation Forest, One-class SVM, etc.).



Data pre-processing

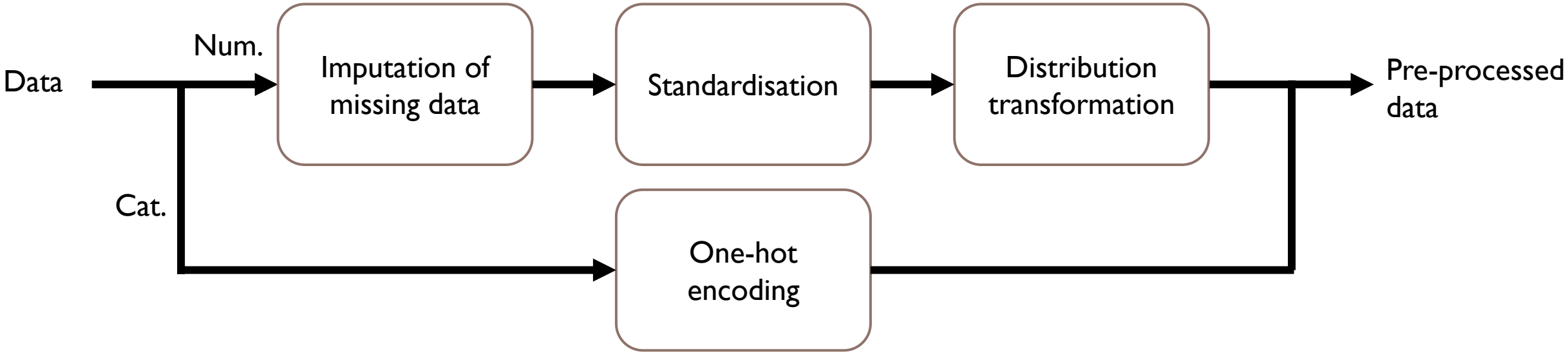


- Making the data suitable for training.

- Data pre-processing may include:
  - Converting text and categorical attributes to numerical attributes (e.g., one-hot encoding, embedding)
  - Scaling the data (normalisation or standardisation)
  - Applying selected transformations to the data distribution (if not close to Gaussian)
  - Undersampling / oversampling data (if imbalanced)
  - Reducing the number of dimensions
  - Etc.

Transformation pipelines

- In practice, you embed all the cleaning and pre-processing step into a pipeline, which can be reused on data as needed.
- The pipeline is designed **using the training data** (important to avoid data leakage) and can then be used to process test set, as well as new instances.



Splitting your data in train and test sets & Further exploration

Conclusion

ONE DOES NOT SIMPLY
START TRAINING A MODEL ON RAW DATA
imgflip.com